

Heinz von Foerster and Niklas Luhmann: The Cybernetics of Social Systems Theory

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I offer a broad comparison between classical and neocybernetic epistemology and sketch the redescription of the subject/object relation as the system/environment relation. I situate von Foerster and Luhmann on the latter side of this comparison and suggest that their shared commitments to a constructivist epistemology informed von Foerster's approval of Luhmann's reworking of Maturana and Varela's concept of autopoiesis.

As a scholar of literature and science, I came to the work of Niklas Luhmann through information, chaos, and complexity theory—topics associated with the rise of cybernetics in the mid-twentieth century shift of technoscientific interest from energy to information, from thermodynamic systems to informatic and communicational systems. What particularly excited me was Luhmann's way of developing social systems theory within the milieu of cybernetic discourse. It was through Luhmann's work that I came to grasp the importance of von Foerster's. To connect them brings forward the cybernetics of social systems theory at the very place where engineering feedback and computation join the biological, epistemological, and communicative issues of natural systems. This complex bridge between the informatic and the autopoietic—between the physical, the living, and the virtual—needs continual shoring up, demands the maintenance of its autopoiesis. I hope that my work at this nexus can contribute to that process.

I will frame the epistemological issues at hand with some broad historical strokes. Table 1 gives an itinerary of Western thought as centered on the standard dualism of subject and object:

Table 1: Epistemology under the Subject/Object Paradigm

Subject	Object
Mind	Nature
The Knower	The Known
(Kantian) Transcendentalism	Empiricism
Romanticism	Classicism
Idealism	Realism
Prescription	Description
Values	Facts
Arts and Humanities	Natural and Social Sciences

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The first thing to note is the pervasive polarism—an oscillating polarity to be sure: the dualism endemic to Western thought. Depending on one’s particular philosophical commitments, the plus and the minus signs flip back and forth from one side of the dichotomy to the other. Within this scheme, an abstract concept of mind—conceptualized as a single or individual faculty of reason ultimately modeled on the omniscient mind of a transcendent god—stands over and against the multiplicity of the world—typically considered as mindless, or in a related dialect, mechanical. The knower is thought to be detached from the known; the knowing subject submits the object to a scrutiny presumed to be capable of yielding a universally valid account, one that leaves the singular yet universal knower out of account. In reaction against such positivistic assumptions and detached postures, in recent centuries philosophies of mind and so-called “romantic” movements arise to protest so-called “classical” philosophies of nature. But even when discourses of qualitative value are held to transcend the quantification of facts; or when a life force is set apart from the supposedly dead world of matter and energy; or when the arts and humanities increasingly precariously stake their institutional rationales on an adversarial relation to an increasingly dominant cultural hegemony of the natural and social sciences, and laments arise over the intellectual schizophrenias induced by the two cultures, and so forth, and so on; even then, this basic framework is not radically challenged.

What are some of the unstated assumptions informing this classical and romantic, subject and object scheme, and what is missing from it? Consider the bar that runs down the middle of the diagram. It implies that the stated polarities, the conceptual cuts that sort out these opposites, are clean and simple—that the critical cleaver has come down true upon the joints of the world and neatly articulated them. It also implies that the bar is itself an “objective” conception standing apart from the subject that posits it. But is this really the case? And concerning an element missing here, we can ask: where does *sociality*—the notion that knowledge is not in fact an individual but a collective enterprise necessarily based on the communication of and not simply the making of observations—enter into this picture? Doesn’t the bar between mind and nature, knower and known, erase the matter of social differentials in favor of singularities idealized as universalities? The particular nexus of cybernetic thought and systems theory initiated by von Foerster and developed further by Luhmann provides an alternative and, so to speak, more wholesome vision of what knowledge is and how it comes about. It is an alternative that leaves intact the benefits of classical thought styles in those local orbits where they may continue to do good work, but that overcomes their blindness to their own blind spots set in place by their particular forms of distinction and their dubious presumptions of universality.

Twenty years after the end of the Macy conferences, von Foerster was consolidating his turn toward second-order cybernetics. In a cluster of papers written by the mid-1970s, von Foerster catalyzed new thinking about the deeper cognitive implications of *circular causality*. The process of recursion unlocked a range of complex self-referential systems, as second-order cybernetics turned its thinking upon itself. In order to arrive at a general discourse of operational circularity, von Foerster

posited his cybernetics of cybernetics as a scientific subject taking itself as its own object. In contrast, the Western epistemological tradition sketched above is calculated precisely, through the linear parsing of subject and object and the insistence on the strict ontological difference between them, to eliminate self-referential circularities. In orthodox Western philosophy, one could say that self-reference as an engine of paradox is granted phenomenological but not properly logical status: it is admitted as a possibility, but only as a possibility of subjective aberration.

Viewed in this wider context, Luhmann's social systems theory extends directly from von Foerster's contributions to the recuperation of self-reference by tracing the ways that, for most a century, hard scientific thought has been forced to acknowledge the paradoxes of observation. Luhmann's (2002) "Cognitive Program of Constructivism" follows von Foerster's lead in tracing the epistemological repercussions of introducing observational self-reference into the knowledge claims of the natural sciences: "Interest in epistemological questions," this essay begins, "is not limited to philosophy today. Numerous empirical sciences have, in the normal course of their research, been forced to proceed from the immediate object of their research to questions involving cognition. Quantum physics is perhaps the best known example, but it is no exception" (Luhmann, p. 128). Von Foerster and Luhmann both push the point that, especially in the rigorous forms demanded by hard science, the production of knowledge on the basis of controlled observations can no longer be conceptualized as a simple or transparent matter. Controlled observations must impose a cognitive construction upon the nature observed, and this state of affairs cannot be eliminated, it can only be compensated for.

One goal of second-order cybernetics, then, has been to lobby scientific constituencies for greater representation of neocybernetic thinking about systemic circularity. For instance, in a 1994 interview, von Foerster discussed the impasses reached in the mainstream research programs of Artificial Intelligence. By disowning their cybernetic origins and consigning feedback and circular mechanisms to a minor role, they had become "nothing but control theory applied to living organisms" (von Foerster, 1994). He expressed a related impatience with the philosophical hangover, epitomized in modern times by the theory of types Russell and Whitehead introduced in *Principia Mathematica*, sweeping the logical circularities of self-reference and paradox under the theoretical rug. Instead,

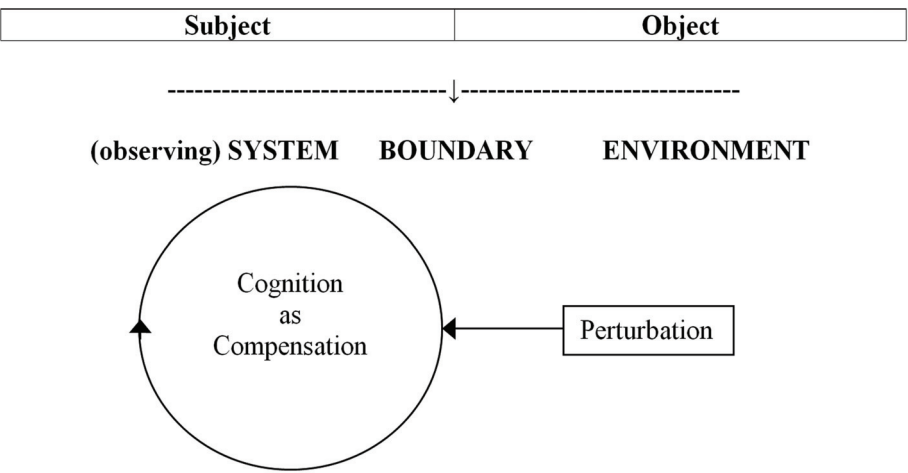
Cybernetics considers systems with some kind of closure, systems that act on themselves—something which, from a logical point of view, always leads to paradoxes since you encounter the phenomena of self-reference. I thought that cybernetics was trying to bring into view a crucial point in logical theory—a point that is traditionally avoided by logic; one which Russell tried to eliminate by bringing in the theory of types. ...

.... Cybernetics, for me, is the point where you can overcome Russell's theory of types by taking a proper approach to the notions of paradox, self-reference, etc., something that transfers the whole notion of ontology—how things are—to an ontogenesis—how things become. ... It is the dynamics of cybernetics that overcomes the paradox. The paradox produces a "yes" when it says a "no," and then produces a "no" from a "yes." It's always a production. In cybernetics you learn that paradox is

not bad for you, but it is good for you, if you take the dynamics of the paradox seriously. (von Foerster, 1994)

There is perhaps no faster way to an encounter with the sort of paradox von Foerster was talking about than by forcing the epistemological issue of second-order systems theory. Luhmann puts the paradox point blank in “The Cognitive Program of Constructivism”: “it is only non-knowing systems that can know; or, one can only see because one cannot see” (Luhmann, 2002, p. 132). Indeed, how can a discourse of knowledge founded on a concept of self-reference, which implies the operational closure of subjects of knowledge reconceptualized as observing systems cut out from their surrounding environments, result in anything but a short circuit or an infinite regress? Let us seek an answer to this issue by clarifying what is at stake in von Foerster’s conceptual shift. Figure 1 offers a diagram that situates the second-order shift within the discourse of general systems theory:

Figure 1: Epistemology Under the System/Environment Paradigm



This cognitive regime bars any traditional form of empirical or realist representationalism, any simplistic notion of knowledge as a mechanics of linear inputs and outputs. Cognized as a system, subjectivity is rendered as a contingent operational effect rather than assumed as a free-floating or even disembodied agency: this is one of the things von Foerster means by stressing, “It’s always a production.” The boundary between subject and object is similarly retrieved from conceptual neutrality and re-cognized as both an ongoing product of and an impassable limit to the operation of the system. A system boundary never just is, ontologically, but is always coming into being as part and parcel of the system’s total ontogenesis, or as this will come to be called, *autopoiesis*. As a self-referential product of the system’s operational enclosure, the boundary guarantees that the system is autonomous, or information-tight. The environment as object cannot enter into the system in the mode

of its own being, cannot dictate to the system. What it can do is perturb its observer in such a way that the system reorganizes its own elements to compensate, which compensation must then count for the system's cognition of the object.

When I interviewed von Foerster, I was particularly interested in asking him about Luhmann's relation to the work of Maturana and Varela. This topic is also important for assessing the scientific elements of Luhmann's work, because of the controversy Luhmann generated by displacing the theory of autopoiesis from the "harder" science of living systems to the "softer" science of social systems. Although this move had been tried before, no one prior to Luhmann had extended autopoiesis in such a complex and coherent way into the metabiotic nexus. Von Foerster, I think, may be said to have agreed, at least implicitly, at least with Luhmann's opening move, that it is not improper to generalize the concept of biological autopoiesis to metabiotic systems as well:

Luhmann himself corresponded a lot with me. He visited me; he was here himself, sitting on this chair; and he was very interested in me because he knew Heinz understands autopoiesis. And he wanted to articulate autopoiesis for social theory. And I always told him, "Be very careful, you will step on Maturana's toes. The most sensitive things, maybe there are already some swellings on his toes." But I know Maturana very well. In fact I was present when the two guys, Varela and Maturana, invented the notion of autopoiesis and wrote their first paper ... Their problem with Luhmann is they do not want to have autopoiesis applied to social theory. This is just a personal idiosyncrasy. Because they would like to keep autopoiesis solely for biological discussion or biological research. (Clarke, 2009, pp. 28-29)

On the basis of this testimony we could speculate that Luhmann corresponded with von Foerster so assiduously, and referenced him so consistently in his published work, not only because of the broad congruence of their conceptual orientations, but also because von Foerster provided Luhmann with an important sign of validation for his critical theoretical leap from biological to social and psychic autopoiesis.

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